

Half-perimeter wirelength

Half-perimeter wirelength is the teaching yardstick for placement

The idea

- For every net
- Sum over nets for total HPWL
- Never celebrate a tiny total that piles every cell on one point

Pseudocode

- HPWL is one loop over nets: bounding-box width plus height, then sum
- That sum is the teaching yardstick for every later placer
- Open this module's examples file and find the Pseudocode section
- That written sketch is what you implement on the implement track and what the browser

Algorithm sketch

- Starter placement scores fifty-two
- Compact golden scores fourteen
- Never celebrate a tiny total that piles every cell on one point

Algorithm sketch — try these

INPUT: positions, nets

OUTPUT: total HPWL

for each net: bbox $\rightarrow$ (maxx-minx)+(maxy-miny)

total $\leftarrow \sum$ net HPWL

GOLDEN starter=52; compact=14

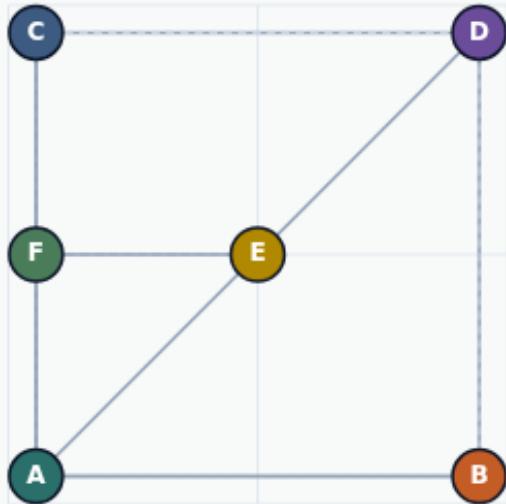
NOTE: collapsed point $\neq$ usable place

Starter placement is spread out

HPWL METRICS

Starter placement is spread out

Step 1 / 5 · starter-spread · module01-01-hpwl-metrics



Explanation

Six cells A–F sit on a rough eight-by-eight canvas. The spread-out seed pulls nets long: total half-perimeter wirelength is fifty-two. This is the shared starter for every placement lab.

- Nodes = movable cells
- Dashed boxes = per-net bounding boxes
- HPWL = sum of (width + height) over nets

Total HPWL: 52
Nets: 6 (one 4-pin)
Cells: A–F

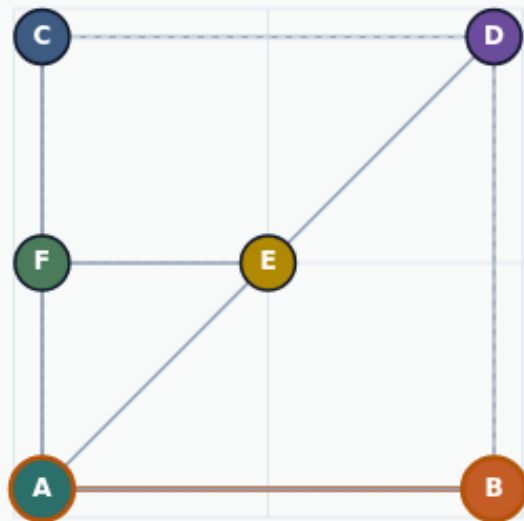
Starter placement is spread out

One net: bbox width plus height

HPWL METRICS

One net: bbox width plus height

Step 2 / 5 · one-net · module01-01-hpwl-metrics



Explanation

Net A–B alone has pins at (0,0) and (8,0). Width is eight, height is zero, so HPWL is eight. Every net uses the same bbox rule before you sum.

- min/max x and y among pins
- $HPWL(net) = (maxX - minX) + (maxY - minY)$
- Two-pin nets reduce to Manhattan span

Net A–B HPWL: 8
Net C–D HPWL: 8
4-pin ABCD HPWL: 16

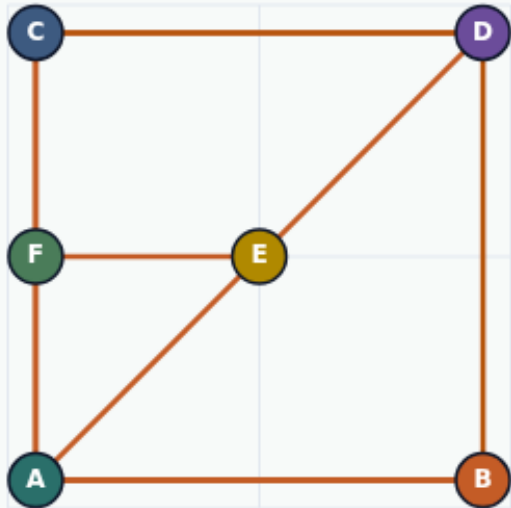
One net: bbox width plus height

Sum six nets to fifty-two

HPWL METRICS

Sum six nets to fifty-two

Step 3 / 5 · sum-nets · module01-01-hpwl-metrics



Explanation

Four two-pin nets at eight each, the four-pin ABCD net at sixteen, and E-F at four: eight times four plus sixteen plus four equals fifty-two. That is the starter golden.

- A-B, C-D, A-C, B-D: 8 each
- ABCD: 16 · E-F: 4
- Never celebrate collapse to one point

$$2\text{-pin} \times 4 = 32$$

$$4\text{-pin} = 16$$

$$E-F = 4$$

$$\text{Total} = 52$$

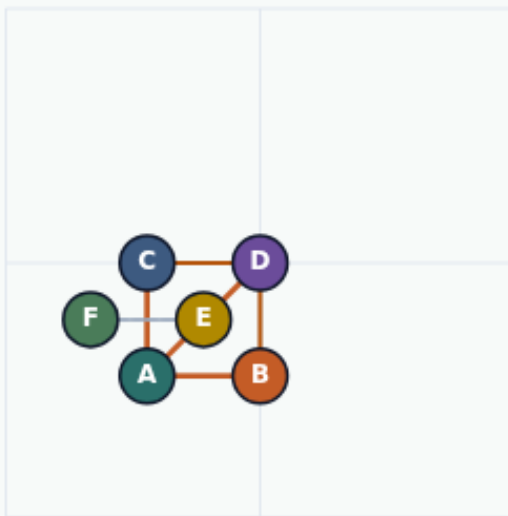
Sum six nets to fifty-two

Golden placement drops to fourteen

HPWL METRICS

Golden placement drops to fourteen

Step 4 / 5 · golden-compact · module01-01-hpwl-metrics



Explanation

The compact reference tucks A–D into a two-by-two block near the center. Same nets, shorter boxes: total HPWL falls from fifty-two to fourteen.

- Same netlist, tighter coordinates
- ABCD bbox shrinks from 16 → 4
- Reference for later algorithms

Starter HPWL: 52

Golden HPWL: 14

Delta: -38

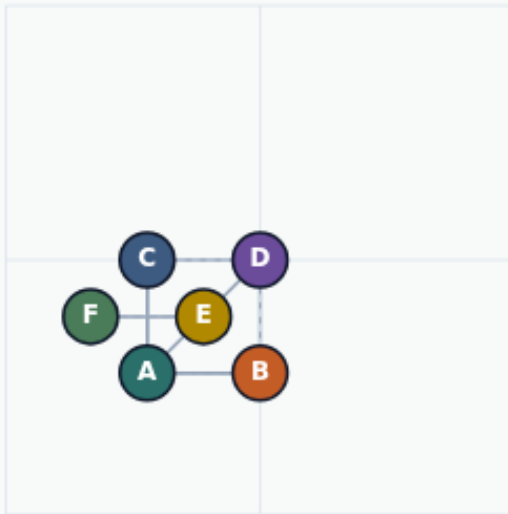
Golden placement drops to fourteen

HPWL is the teaching yardstick

HPWL METRICS

HPWL is the teaching yardstick

Step 5 / 5 · takeaway · module01-01-hpwl-metrics



Explanation

Report total HPWL before and after every move. Fifty-two versus fourteen is the starter story—but a tiny total with stacked cells is not a usable placement.

- Compute bbox HPWL per net, then sum
- Starter 52 · golden 14
- Next: clique and star models

```
starterHpwl: 52  
goldenHpwl: 14
```

HPWL is the teaching yardstick

Browser lab track

- In the browser lab track, open the **hpwl-metrics** lab from the tools shelf
- Load the starter placement, run the algorithm once
- Work the challenges that lock the goldens

Implement track

- In the implement track
- Parse ``tiny_place.json``, run the algorithm with a deterministic seed
- Match the browser goldens before you claim the checklist

Pitfalls

- Common traps

Your turn

- Complete the checklist for at least one track, preferably both
- Implement until your metrics match the starter goldens
- When you're ready, take the short quiz, then continue to the next module